

MAKAUT 2025 - Machine Learning Solved Answer Paper

Paper Code: MCAN-305G | Subject: Machine Learning

Full Marks: 70 | Time: 3 Hours

Group - A (Very Short Answer Type Questions) (1 mark each)

1. Overfitting: Model fits noise and fails to generalize to new data. (1 mark)
2. Gradient Descent: Optimization algorithm minimizing cost by adjusting weights. (1 mark)
3. Decision Tree Category: Belongs to supervised learning. (1 mark)
4. ANN: Artificial Neural Network. (1 mark)
5. PCA: Used for dimensionality reduction. (1 mark)
6. Bagging vs Boosting: Bagging reduces variance; Boosting reduces bias. (1 mark)
7. Conditional Probability: $P(A|B) = P(A \text{ and } B) / P(B)$. (1 mark)
8. Kernel (SVM): Function to transform data into higher-dimensional space. (1 mark)
9. Activation Functions: Sigmoid, ReLU, Tanh. (1 mark)
10. Logistic Regression: Used for binary classification. (1 mark)
11. Entropy: Measure of impurity in a dataset. (1 mark)
12. Normalization: Scaling features to uniform range. (1 mark)
13. Clustering Algorithm Example: K-Means. (1 mark)
14. SVM is discriminative classifier: True. (1 mark)
15. Measure of randomness: Entropy. (1 mark)

Group - B (Short Answer Type Questions) (5 marks each)

1. Classification vs Regression:
 - Classification predicts discrete labels.
 - Regression predicts continuous numeric values.
 - Both are supervised learning.
 - Example: Spam detection vs price prediction.
 - Output: Classification = category, Regression = value. (5 marks)
2. Naive Bayes Classifier:
 - Based on Bayes Theorem assuming feature independence.
 - Formula: $P(C|X) = [P(X|C) \cdot P(C)] / P(X)$.
 - Calculates probability for each class.
 - Chooses class with highest posterior probability.
 - Example: Email spam detection. (5 marks)
3. Bagging and Boosting:

- Bagging: Parallel models trained on random data subsets (e.g., Random Forest).
- Boosting: Sequential models trained to fix previous errors (e.g., AdaBoost).
- Bagging -> reduces variance.
- Boosting -> reduces bias.
- Both improve accuracy. (5 marks)

4. Gradient Descent:

- Minimizes cost function iteratively.
- Formula: $\theta = \theta - \alpha * dJ/d\theta$.
- Types: Batch, Stochastic, Mini-batch.
- Used in regression, ANN.
- Controlled by learning rate α . (5 marks)

5. K-Means Clustering:

- Unsupervised algorithm dividing data into K clusters.
- Steps: Initialize K centers -> Assign points -> Recalculate centers -> Repeat.
- Stops when centers stabilize.
- Objective: Minimize intra-cluster distance.
- Applications: Image segmentation, grouping users. (5 marks)

6. PCA (Principal Component Analysis):

- Reduces data dimensions while preserving variance.
- Steps: Standardize data -> Covariance matrix -> Eigen decomposition -> Choose top components -> Transform.
- Helps in noise reduction and visualization.
- Used before ML models for better performance. (5 marks)

Group - C (Long Answer Type Questions) (15 marks each)

1. Linear Regression:

- Predicts continuous dependent variable.
- Simple: $Y = a + bX$; Multiple: $Y = a + b_1X_1 + b_2X_2 + \dots + b_nX_n$.
- Correlation measures relation strength.
- Gradient Descent used for parameter tuning.
- Applications: Price, trend prediction. (15 marks)

2. Decision Tree Learning:

- Splits data using best attribute based on Information Gain.
- Entropy = $-\sum(p * \log_2(p))$.

- $IG = Entropy(parent) - \text{Weighted entropy(children)}$.
- Repeat until all nodes pure or min size.
- Pros: Easy to interpret; Cons: Overfitting possible. (15 marks)

3. Artificial Neural Network (ANN):

- Layers: Input -> Hidden -> Output.
- Neuron: $y = f(\text{Sum}(w*x) + b)$.
- Activation: Sigmoid, ReLU, Tanh.
- Learning via Backpropagation using Gradient Descent.
- Used in vision, NLP, and speech tasks. (15 marks)

4. Support Vector Machine (SVM):

- Finds optimal hyperplane separating data classes.
- Maximizes margin between support vectors.
- Equation: $w*x + b = 0$.
- Kernel trick handles non-linear separability.
- Example kernels: Polynomial, RBF. (15 marks)

5. Ensemble Learning:

- Combines multiple models for better performance.
- Bagging (Random Forest): Uses random samples.
- Boosting (AdaBoost): Sequential weak learners.
- Reduces bias and variance.
- Increases prediction stability. (15 marks)

6. Expectation Maximization (EM):

- Iterative algorithm for soft clustering.
- E-step: Estimate probabilities of data points in clusters.
- M-step: Maximize parameters to fit data.
- Repeats until convergence.
- Example: Gaussian Mixture Models. (15 marks)